25901393 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEMATA01 - INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer any **20** questions

Each question carries **2** marks

1. The Turing test checks whether a machine can [K] / [CO1]

   a). Calculate fast b). Store memory

   c). Exhibit intelligent behaviour d). Draw images

2. The Dartmouth conference was held in [K] / [CO1]

   a). 1945 b). 1956

   c). 1965 d). 1975

3. During disillusionment, [K] / [CO1]

   a). Funding rises b). Interest drops

   c). Research stops permanently d). Users increase

4. AI without learning would [K] / [CO1]

   a). Improve b). Remain static

c). Become human d). Self-repair

5. Training deep networks is feasible today due to [K] / [CO1]

a). GPUs b). Paper

c). Pens d). Telephones

6. Overfitting happens when the model [U] / [CO1]

a). Learns too little b). Memorizes training data

c). Deletes data d). Stops training

7. A smartphone performing on-device recognition uses [K] / [CO2]

a). Cloud AI b). Edge AI

c). Super AI d). Manual AI

8. A narrow AI system usually cannot [K] / [CO2]

a). Compute b). Learn outside its training

c). Process input d). Produce output

9. Building AGI is difficult because [K] / [CO2]

a). Computers are slow b). Human intelligence is complex

c). No data exists d). Algorithms are easy

10. A major cloud provider is [K] / [CO2]

a). Amazon Web Services b). Yahoo

c). Internet explorer d). Firefox

11. Watson combined AI with [K] / [CO2]

a). Natural language processing b). Plumbing

c). Mechanics d). Graphics

12. Modern AI benefits from open-source [U] / [CO2]

a). Barriers b). Collaboration

c). Restrictions d). Isolation

13. Running AI on edge helps when [K] / [CO2]

a). Internet is slow b). Power is unlimited

c). Data is small d). Storage is high

14. Which tool is widely used for data visualization in Python? [K] / [CO3]

a). Matplotlib b). Notepad

c). BIOS d). Router

15. np.arange(0,5) produces [A] / [CO3]

a). [0,1,2,3,4] b). [0,1,2,3,4,5]

c). [1,2,3,4,5] d). Error

16. Which hardware is most commonly used to accelerate deep learning? [K] / [CO3]

a). HDD b). CPU

c). GPU d). Printer

17. Python’s simplicity helps beginners because [K] / [CO3]

a). It removes mathematics b). Syntax is closer to natural language

c). No libraries exist d). Programs are shorter only

18. Data preprocessing usually involves [A] / [CO3]

a). Cleaning and transforming data b). Deleting models

c). Compiling code d). Installing OS

19. Which statement is TRUE about tuples? [K] / [CO3]

a). They can change size b). They are mutable

c). They are immutable d). They store only numbers

20. Which function is used to receive user input? [K] / [CO3]

a). get() b). read()

c). input() d). scan()

21. Normalization is used to [K] / [CO4]

a). delete rows b). scale data

c). color graphs d). create arrays

22. Neural networks are powerful for [U] / [CO4]

a). simple arithmetic only b). pattern recognition

c). typing d). formatting

23. Overfitting occurs when the model [U] / [CO4]

a). performs well on training but poorly on new data b). runs fast

c). uses less memory d). is simple

24. Overfitting means [K] / [CO4]

a). good generalization b). poor performance on unseen data

c). faster execution d). fewer features

25. Missing values in pandas are often represented as [U] / [CO4]

a). 0 b). NA / NaN

c). -1 d). blank

[2x20 = 40]

Part B

Essays

Answer any **3** questions

Each question carries **10** marks

26. Distinguish between AI and Data Science. Explain the Gartner Hype Cycle in AI development. [U] / [CO1]

27. Discuss the future of large AI models. Discuss applications of GPT models. [K] / [CO2]

28. What is Super AI? Discuss possible impacts of Super AI. [U] / [CO2]

29. What is data visualization? Write Python code to plot sine and cosine curves. [K] / [CO3]

30. Explain unsupervised learning with examples. [U] / [CO4]

[10x3 = 30]